

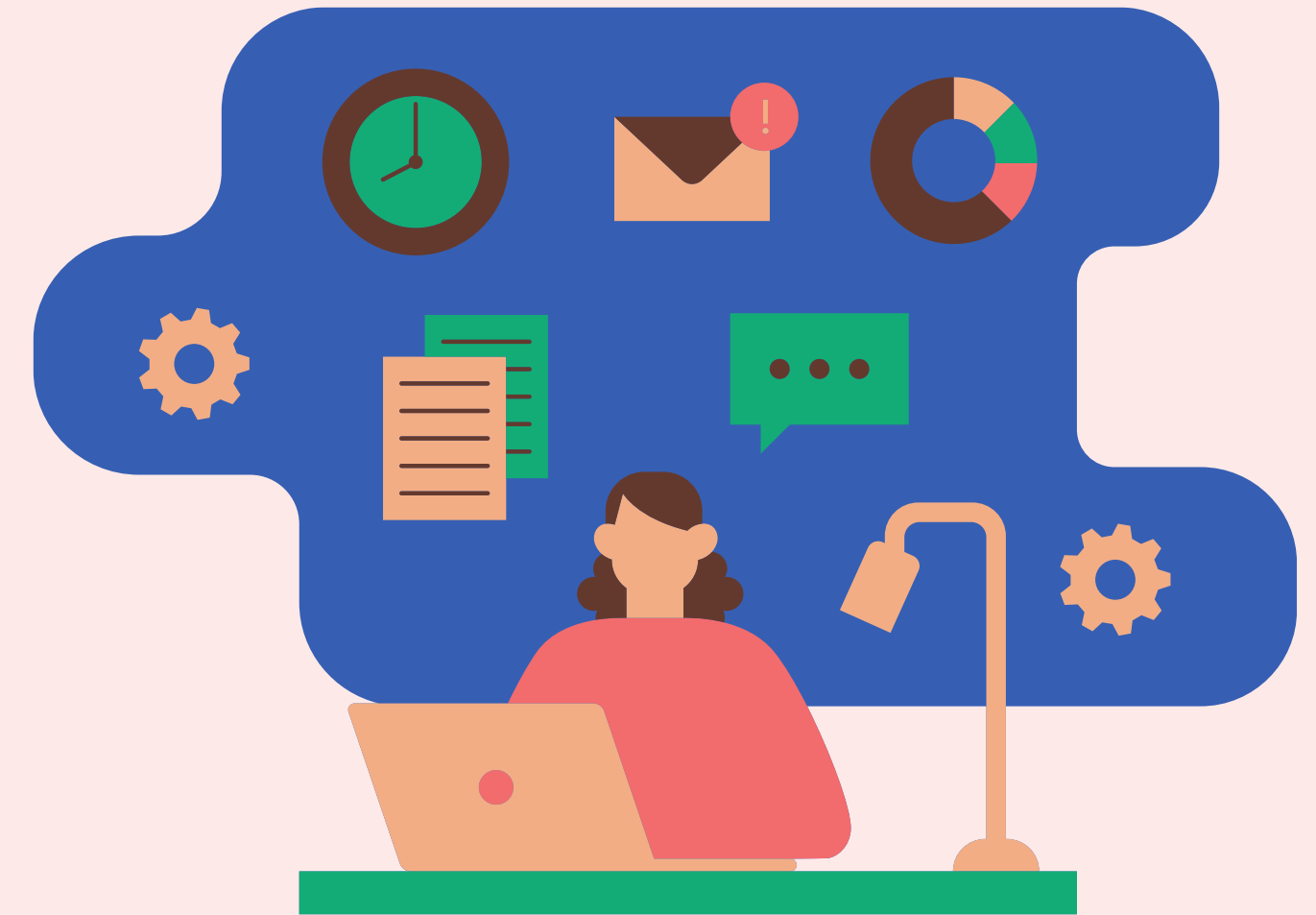
INDUSTRIAL PROJECT MANAGEMENT

5TH SEMESTER SESSIONAL PRESENTATION

PROJECT TITLE:

AGILE IMPLEMENTATION FOR A PIZZA DELIVERY APP “*PIZZAMANIA*”

Presentation by -
3rd Yr CSE/ Section-C Gr. C6/ Team No. 2



TEAM MEMBERS:

- Patatri Acharya
- Roushan Paul
- Koyelisha Khan
- Alivia Nandi
- Adrija Ghosh
- Shyantana Kundu
- Debarati Das
- Tideesha Saha

Project Overview

PROJECT TITLE: AGILE IMPLEMENTATION FOR A PIZZA DELIVERY APP “*PIZZAMANIA*”
CLIENT PROFILE:

- Business Overview: Web-based **High-quality pizza delivery**, **Quickly** and **Reliably** nationwide.
- Vision: Bring great pizzas to all locations and ensuring customer satisfaction.
- Collaboration: With local restaurants and **Delivery points**, creating sustainable ecosystem & support local businesses.

OBJECTIVES:

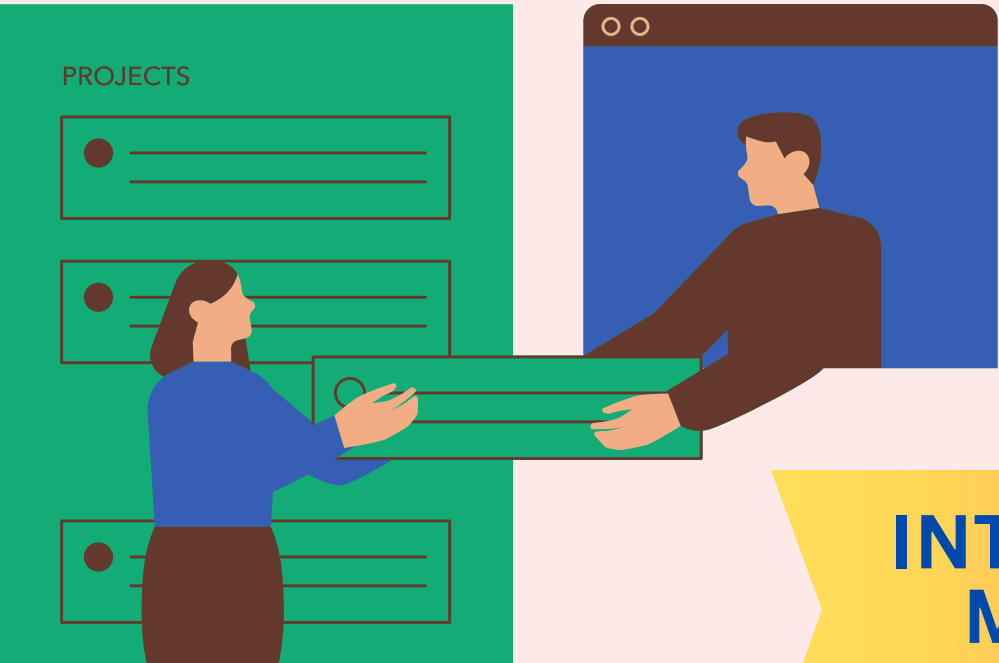
ENGAGING
HOMEPAGE

COMPREHENSIVE
ADMIN CONTROLS

STREAMLINED EDITING &
CHECKOUT OPTIONS

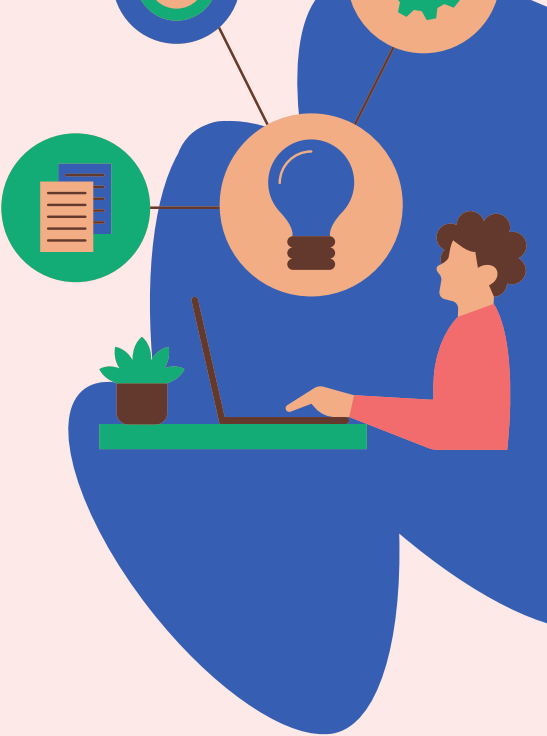
INTUITIVE
MENU
DISPLAY

RELIABLE
DELIVERY
TRACKING

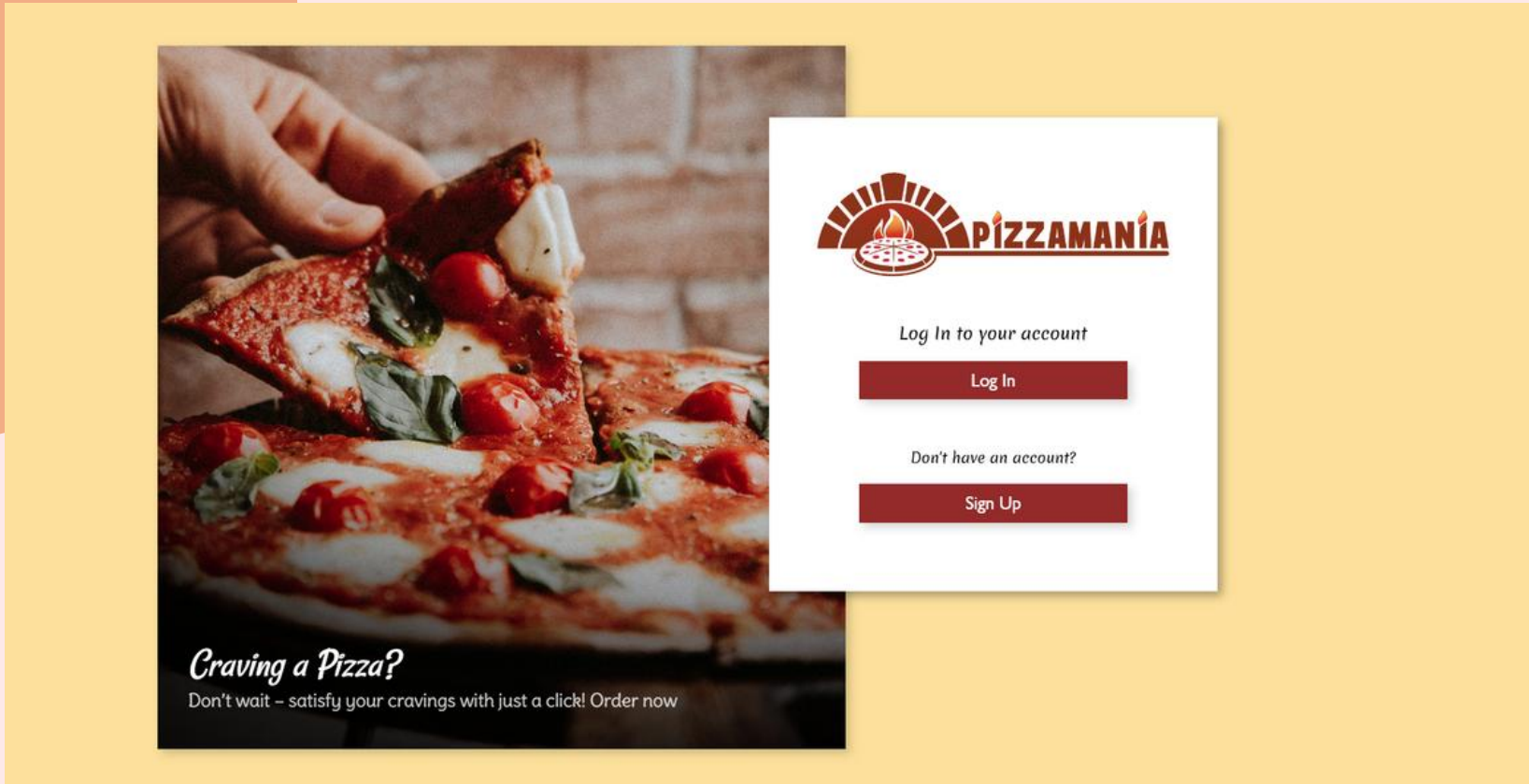


SCRUM TEAM MEMBERS:

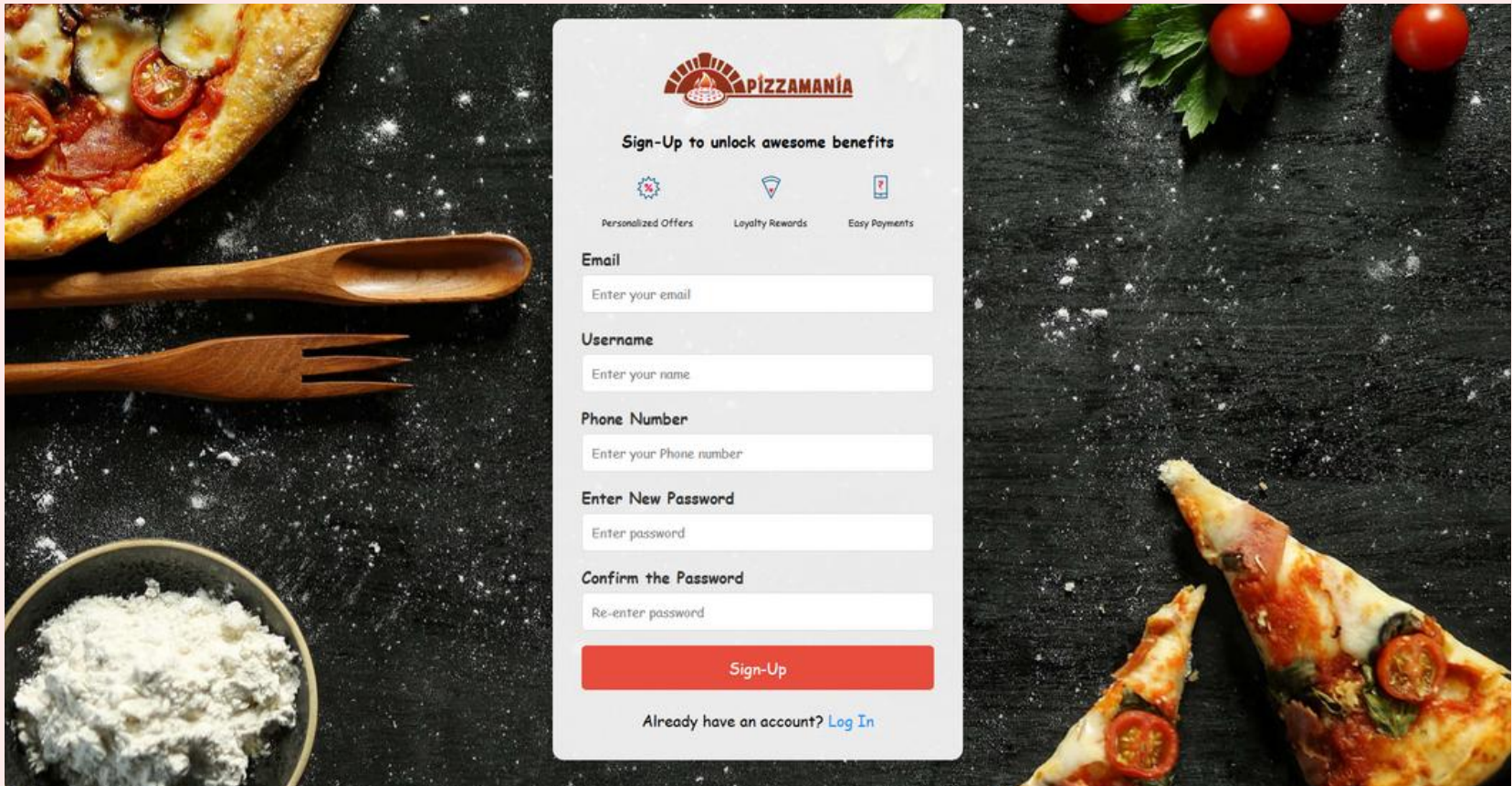
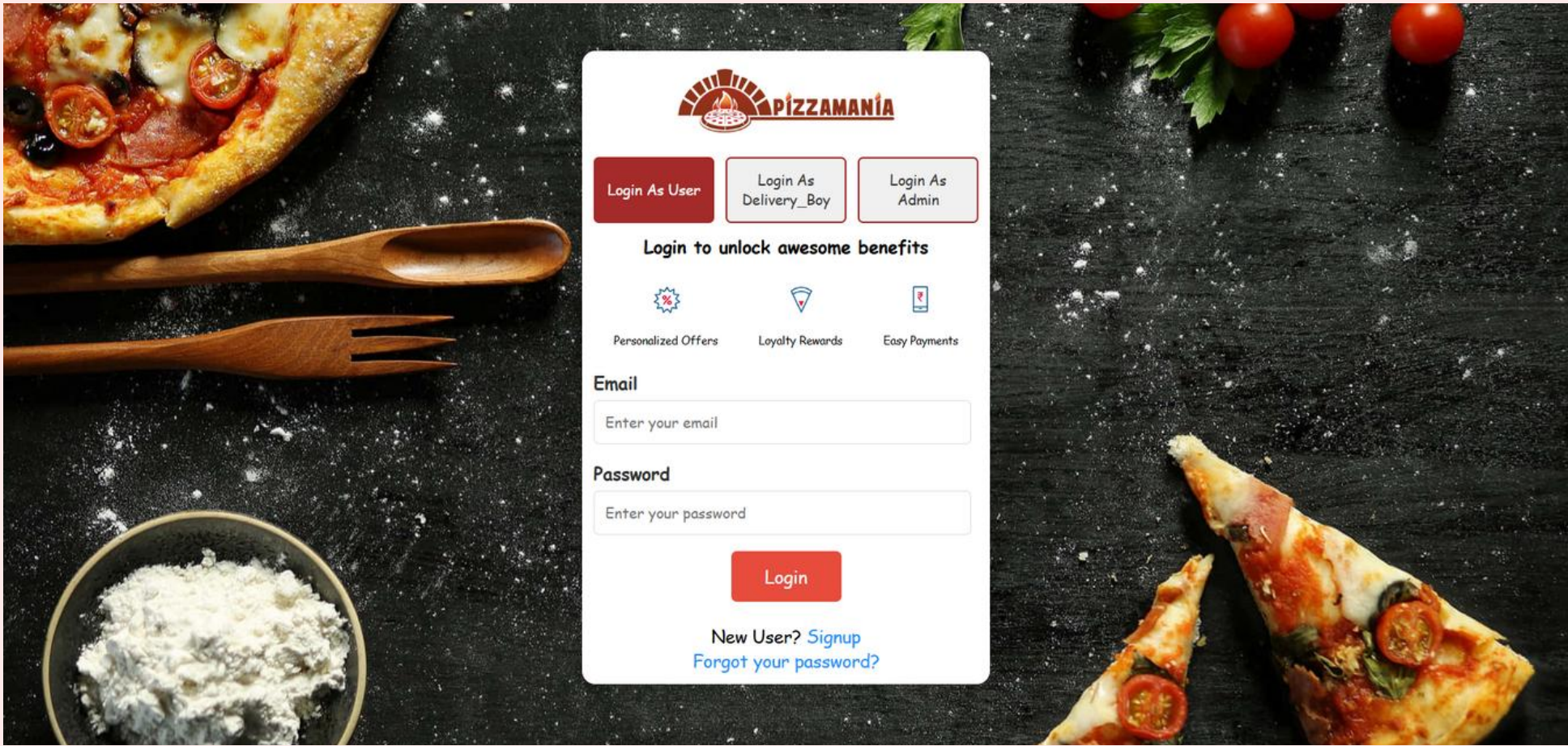
- PRODUCT OWNER:
PATATRI ACHARYA
- SCRUM MASTER:
ROUSHAN PAUL
- DEVELOPMENTAL TEAM:
KOYELISHA KHAN
ALIVIA NANDI
ADRIJA GHOSH
SHYANTAN KUNDU
- TESTING TEAM:
TIDEESHA SAHA
DEBARATI DAS



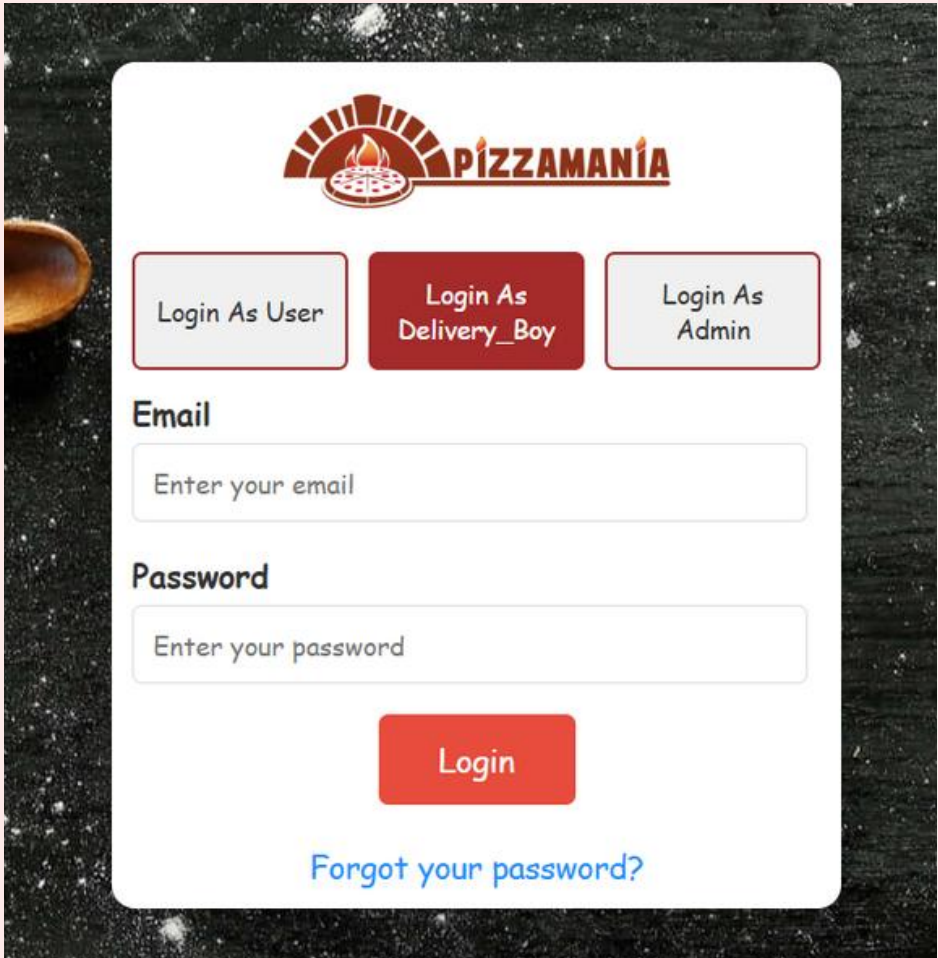
Sprint 1 Deliverables:



Welcome Page

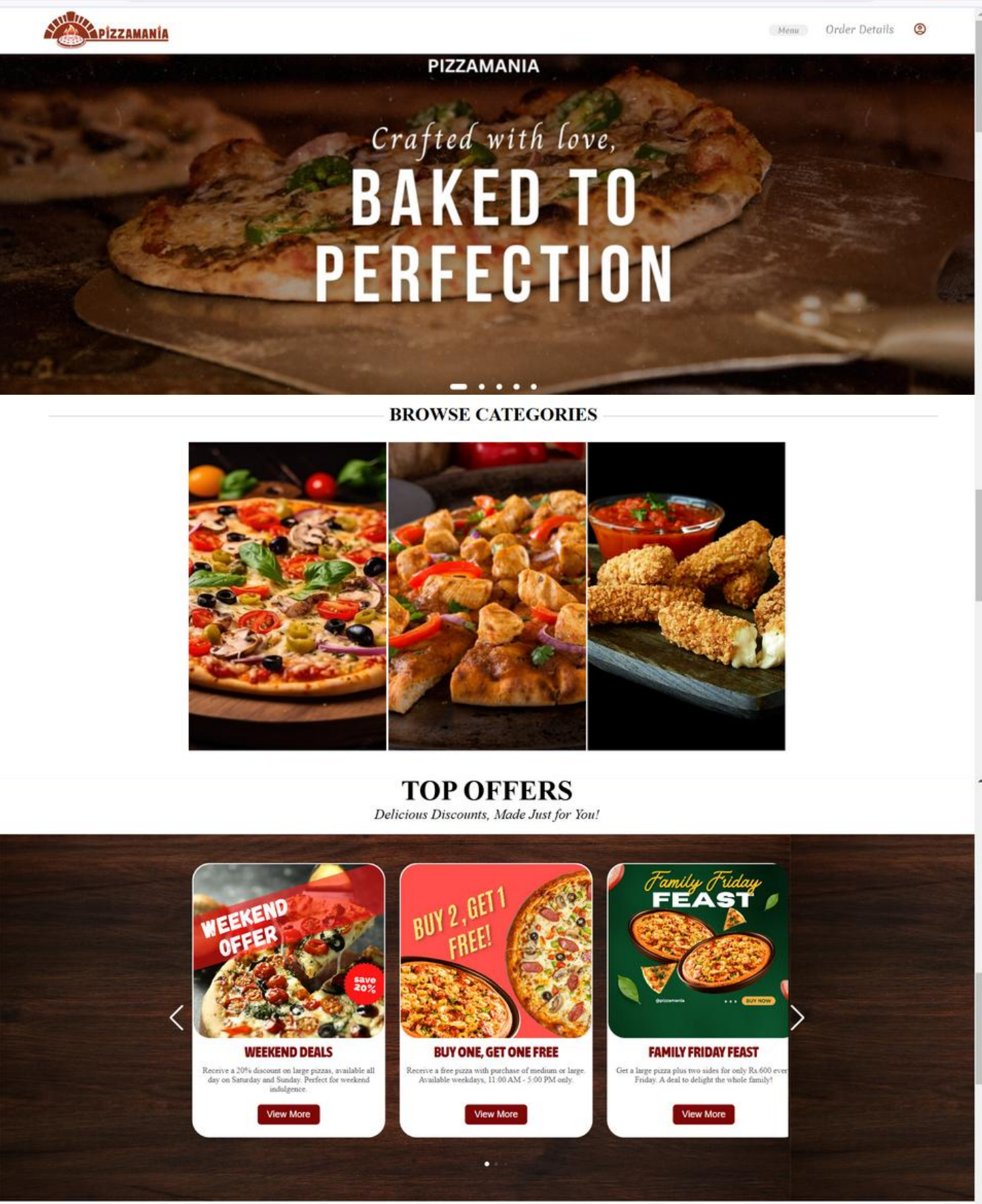


Sign-up page

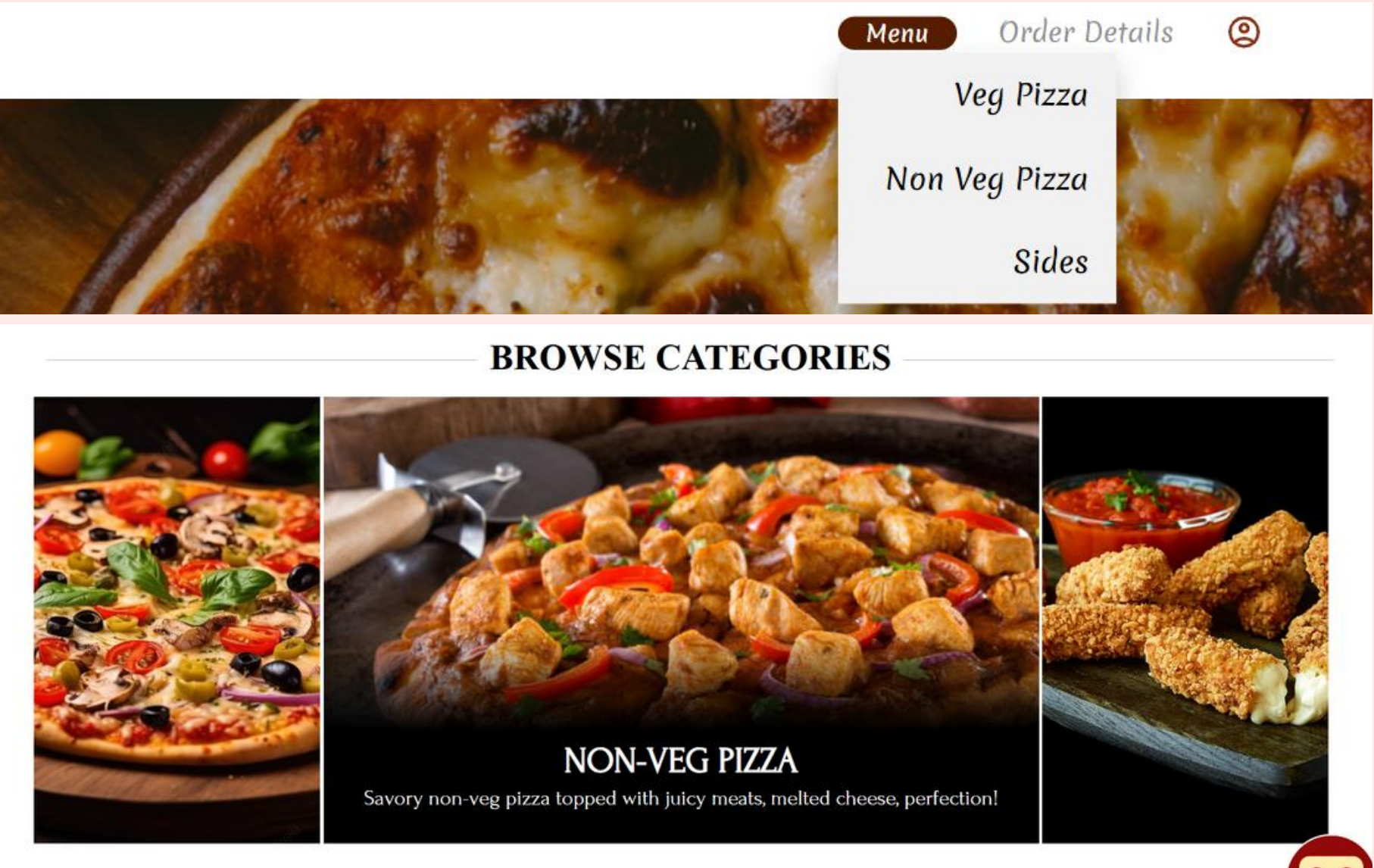


Login pages as User, Delivery Agent & Admin

Sprint 1 Deliverables:



Homepage

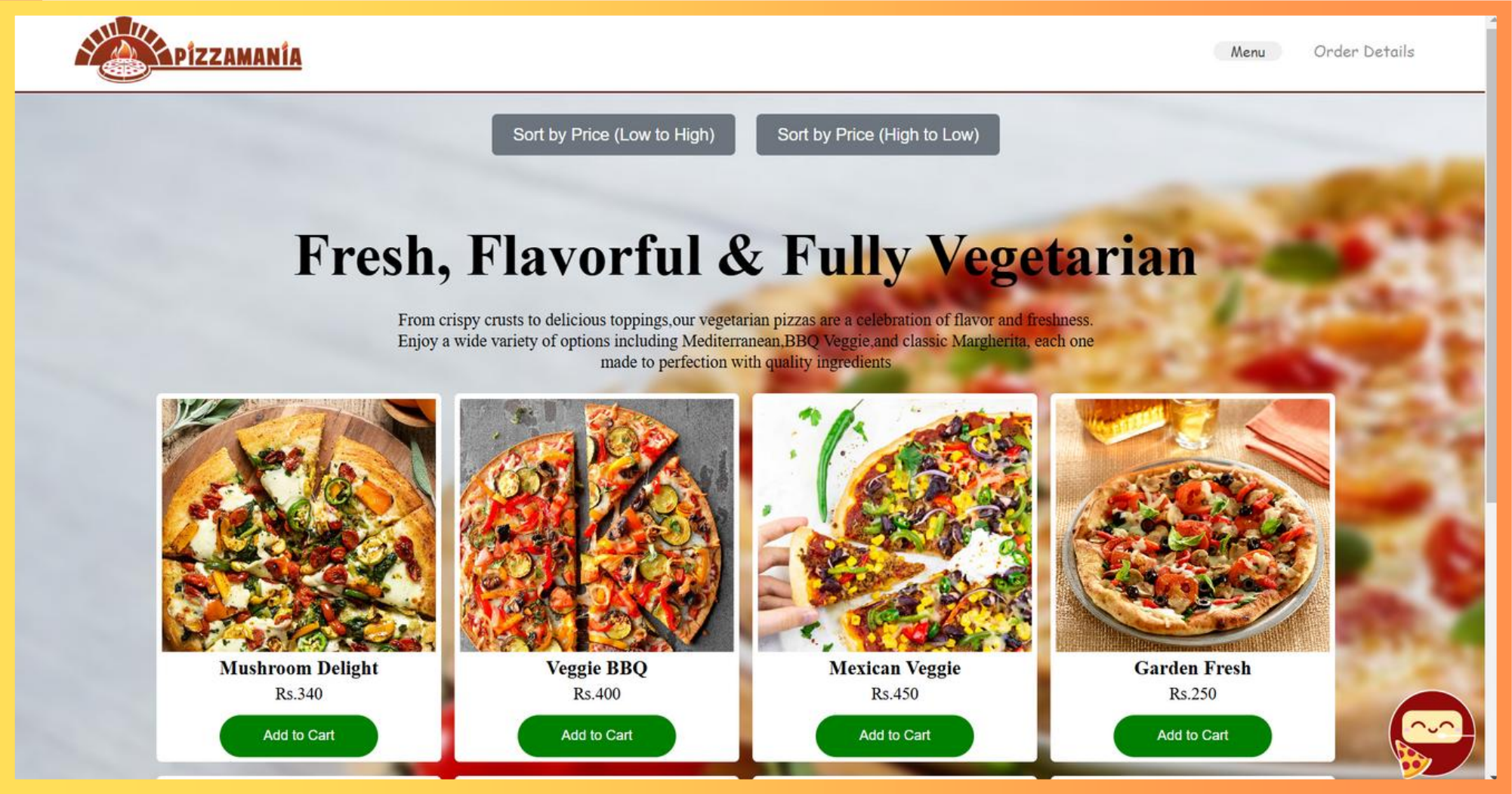


Browsing Menu

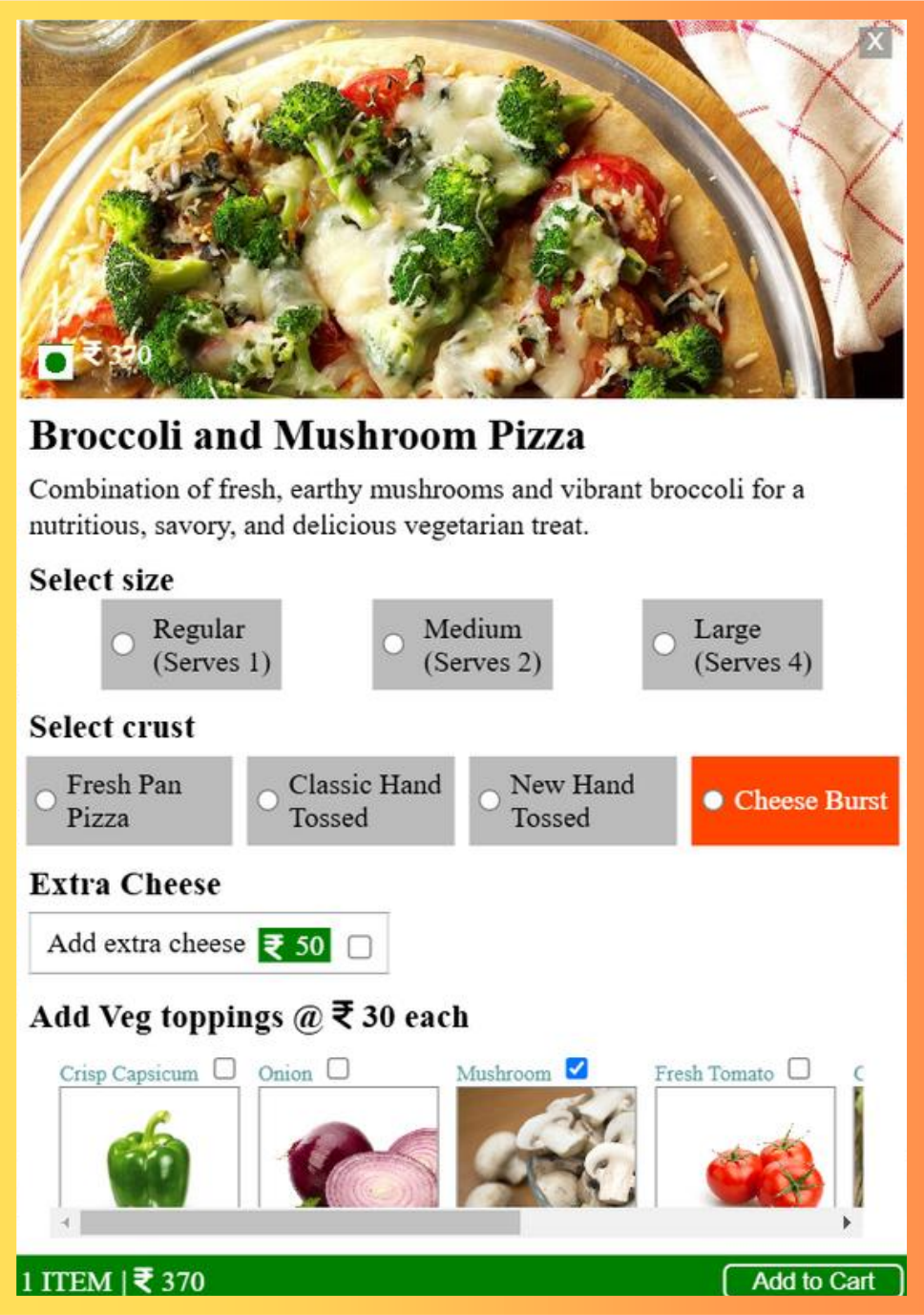
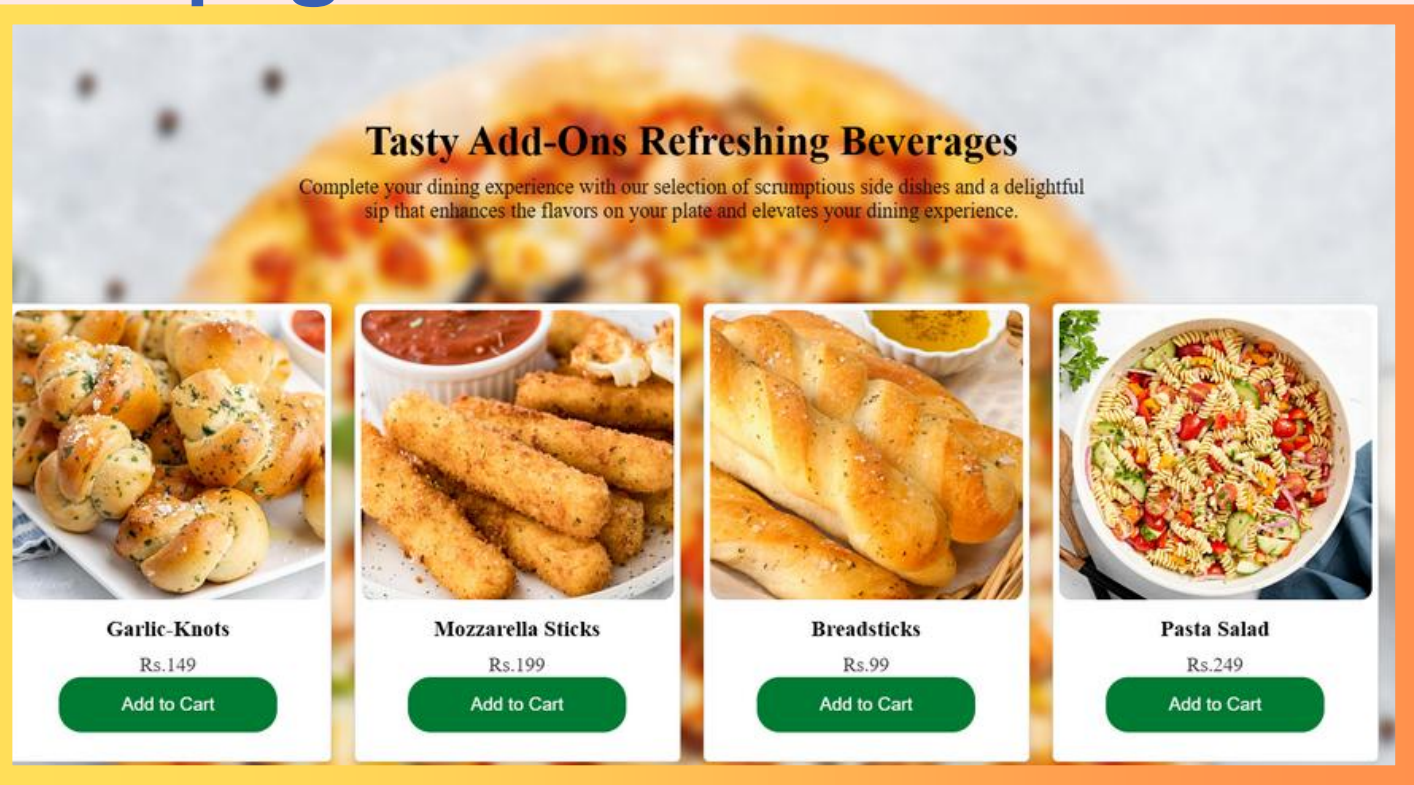
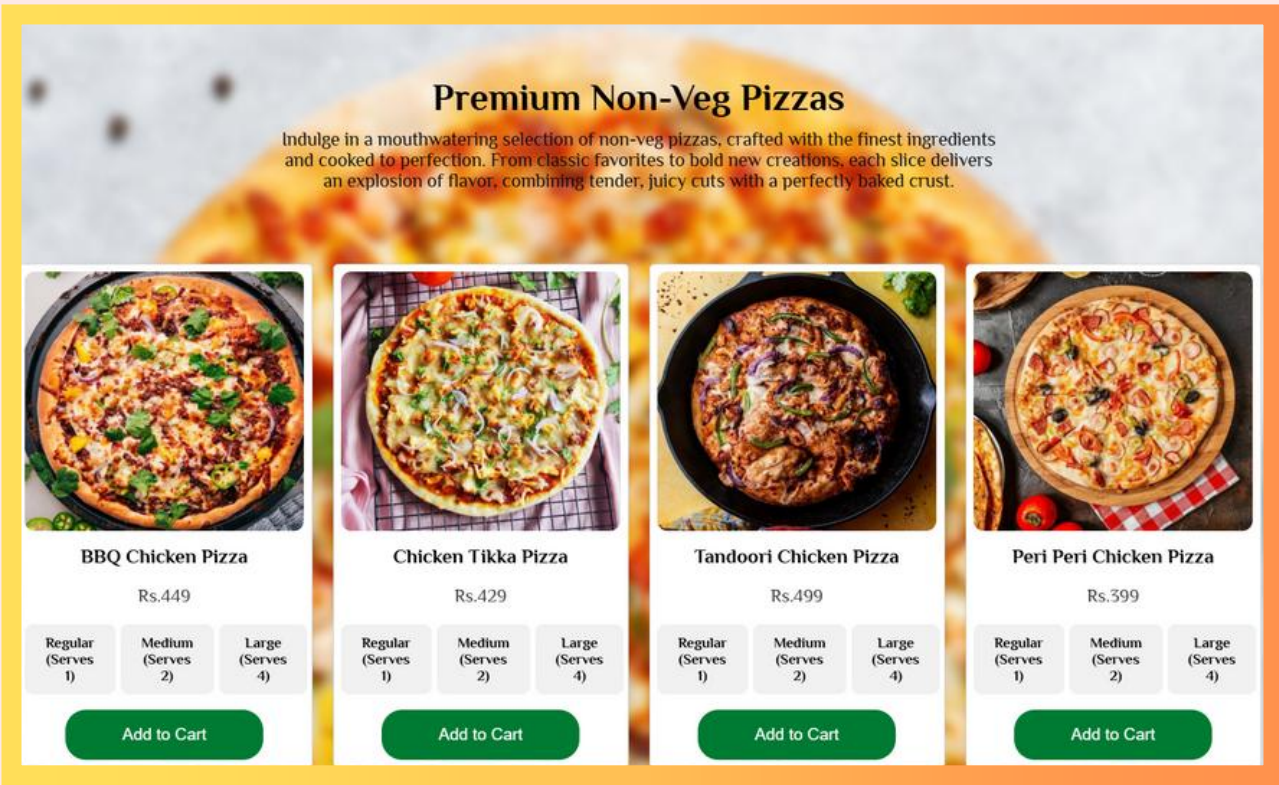


Customer Review Section

Sprint 1 Deliverables:



Product-page



Product customization page

Sprint 1 Deliverables:

```
# Train a KNN model
knn_model = NearestNeighbors(n_neighbors=5, metric='euclidean')
knn_model.fit(features_scaled)

# Define the recommendation function
def recommend_pizzas(pizza_name, knn_model, pizza_data_cleaned, scaler, n_recommendations=5):
    # Locate the pizza in the dataset
    pizza_index = pizza_data_cleaned[pizza_data_cleaned['Name of Pizza'] == pizza_name].index
    if len(pizza_index) == 0:
        return f"Pizza flavor '{pizza_name}' not found in the dataset."

    # Transform the selected pizza's features
    pizza_features = features_scaled[pizza_index[0]].reshape(1, -1)

    # Find the nearest neighbors
    distances, indices = knn_model.kneighbors(pizza_features, n_neighbors=n_recommendations + 1)

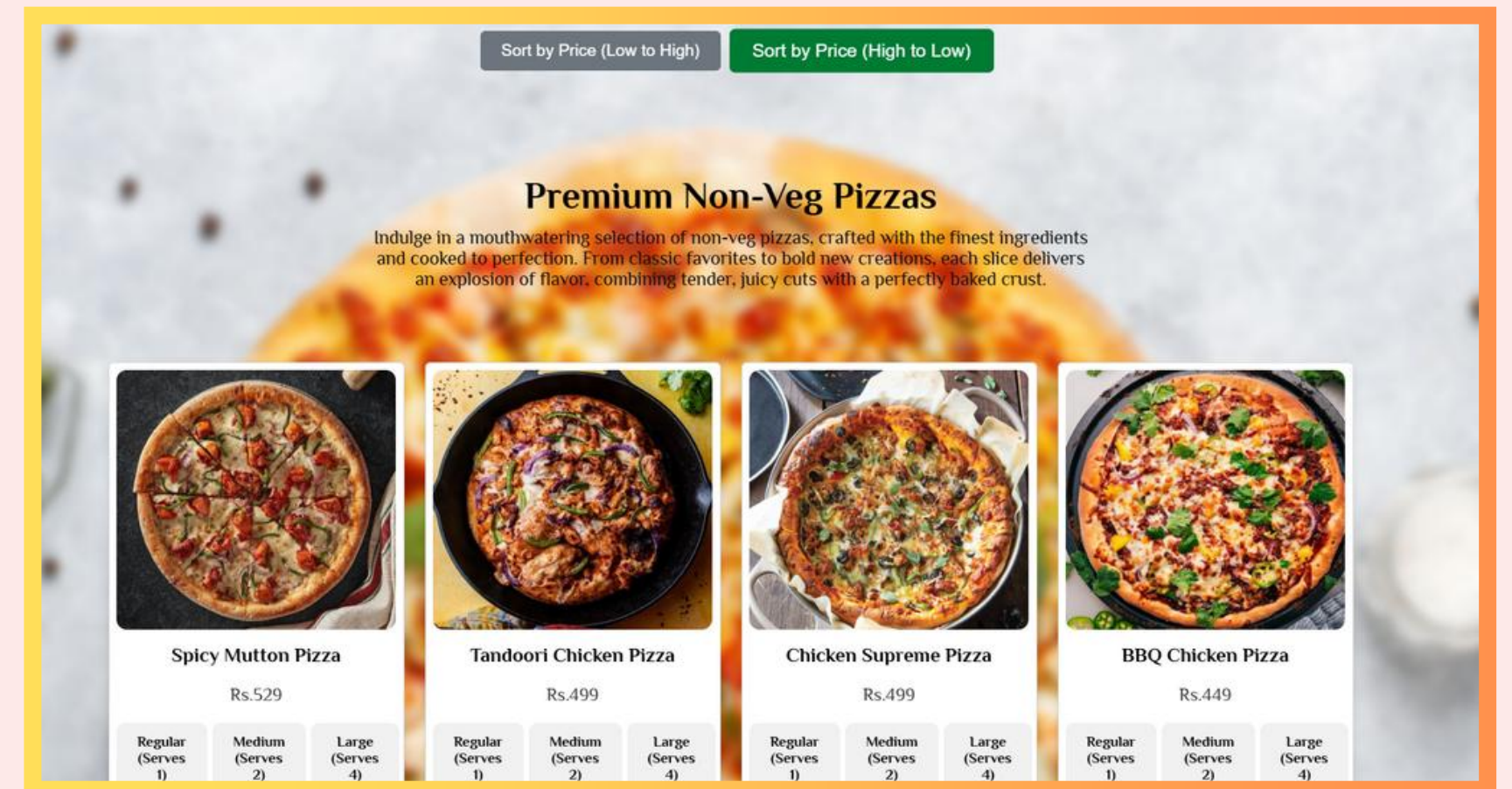
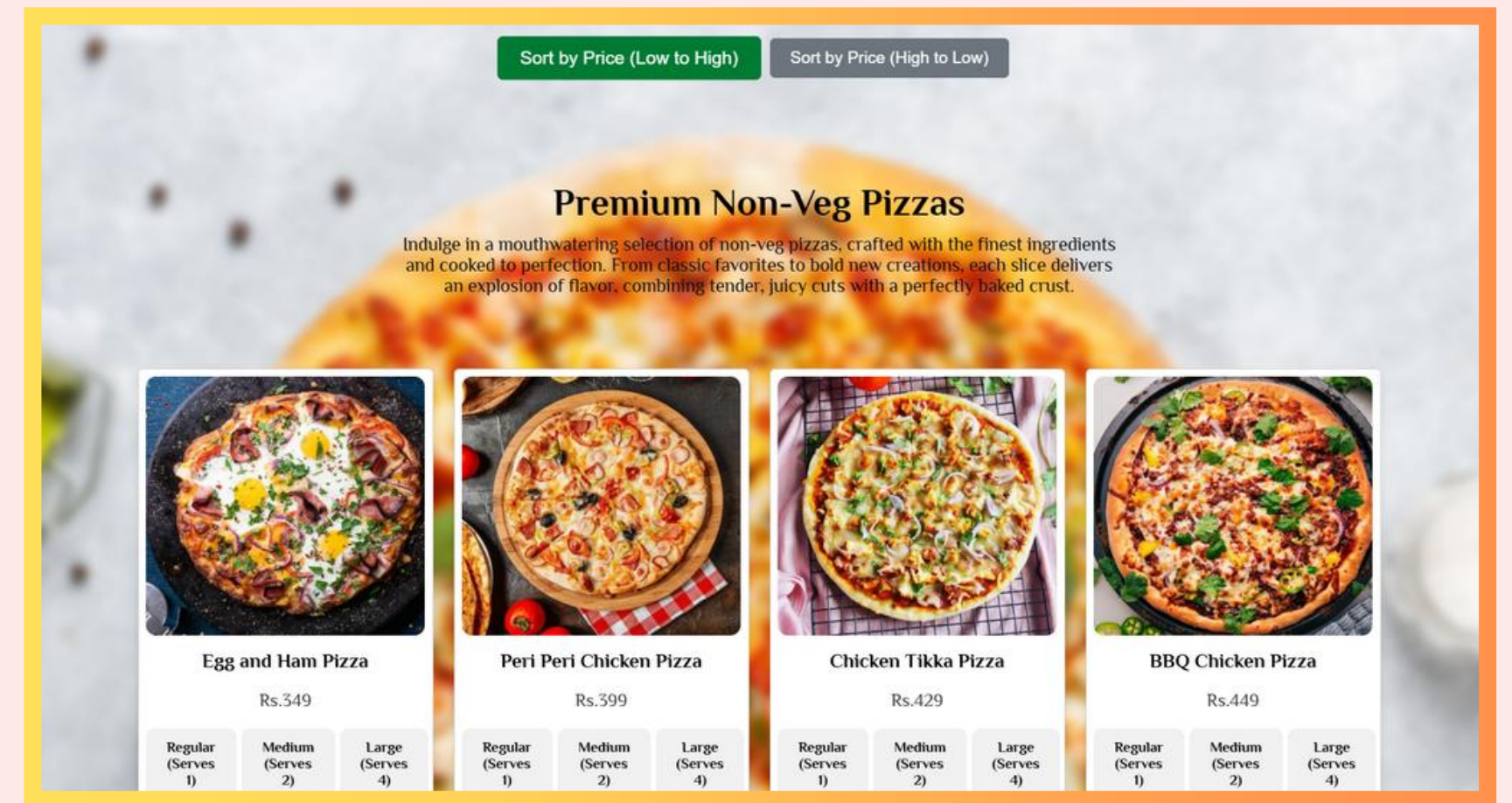
    # Get recommended pizza names, excluding the input pizza
    recommended_indices = indices[0][1:] # skip the first index (input pizza itself)
    recommendations = pizza_data_cleaned.iloc[recommended_indices]['Name of Pizza'].tolist()

    return recommendations
```

```
sample_pizza_name = "Multigrain Pizza"
recommend_pizzas(sample_pizza_name, knn_model, pizza_data_cleaned, scaler)
```

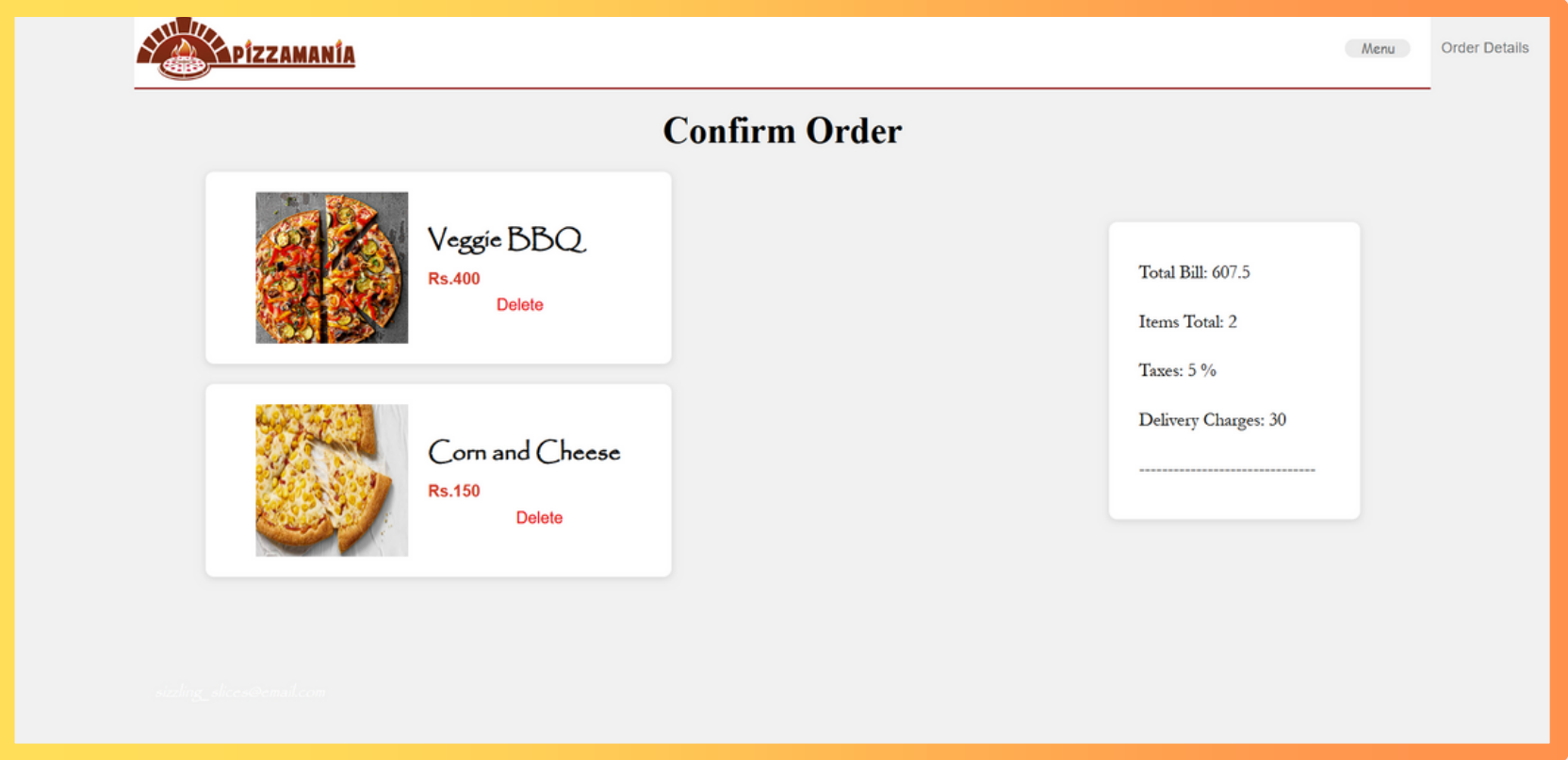
```
['Wholegrain Pizza',
 'Vegetarian Pizza',
 'Margherita Pizza',
 'Pepperoni Pizza',
 'Chicken Supreme Pizza']
```

Pizza-recommendation Model

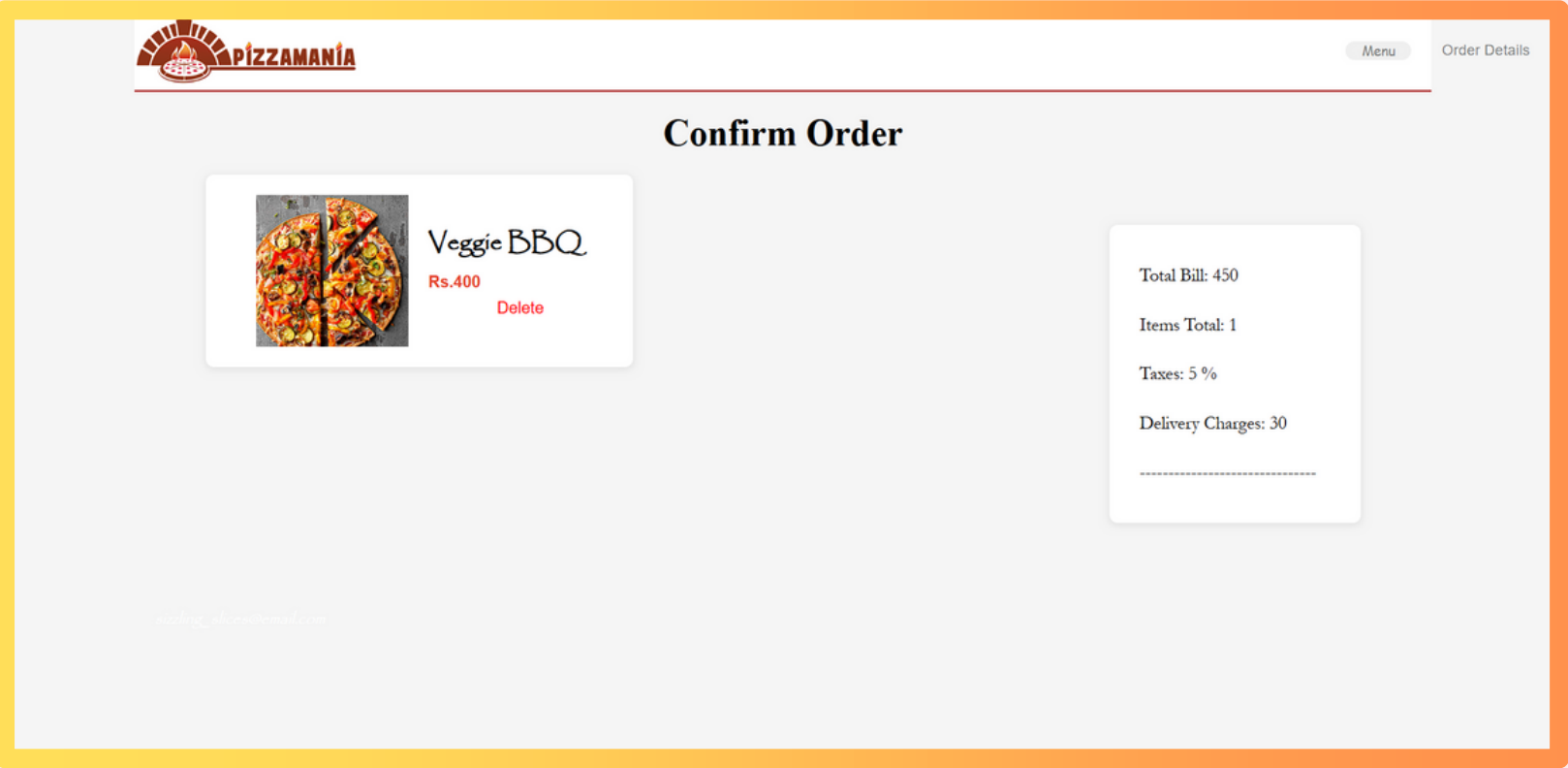


Sort Pizzas by price

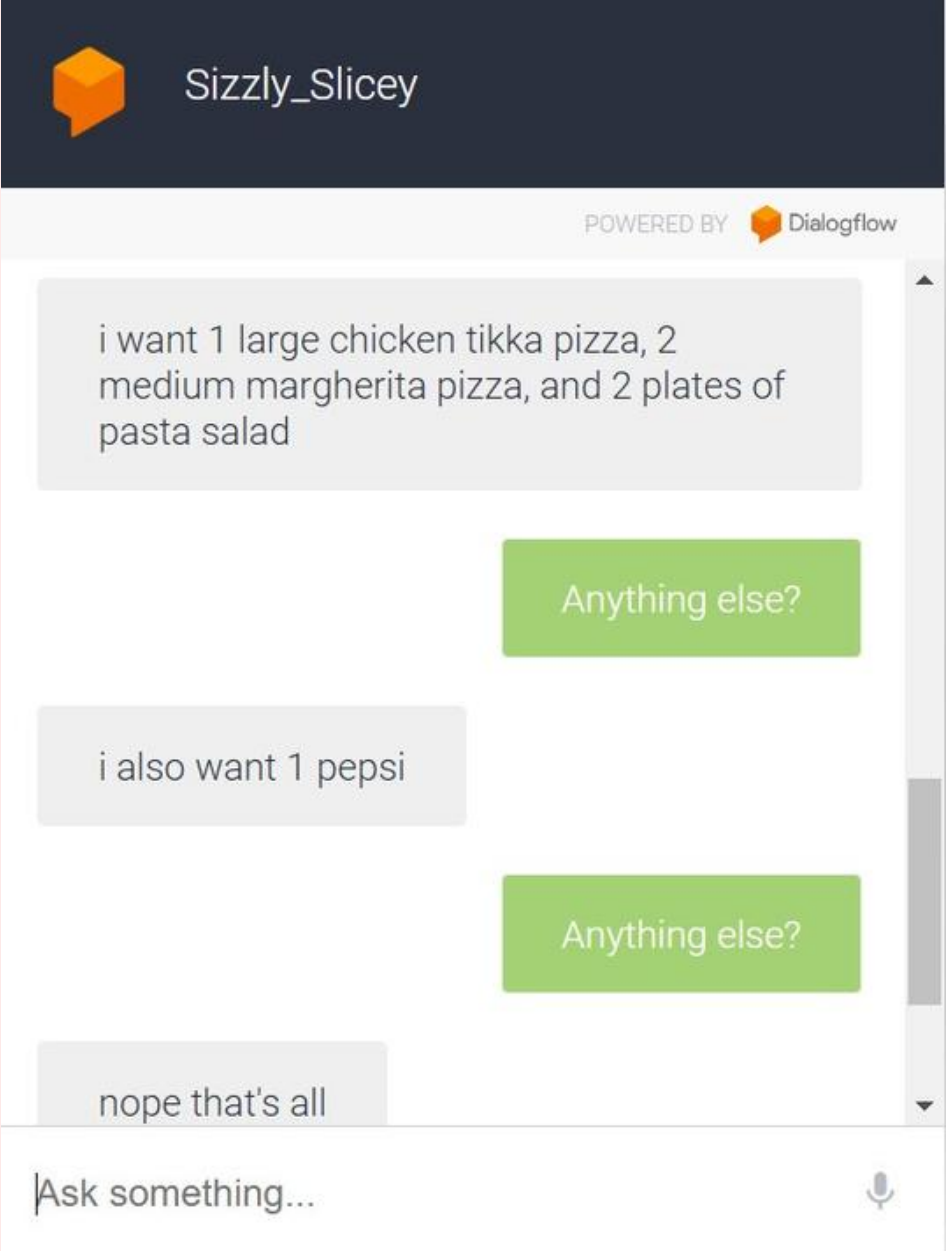
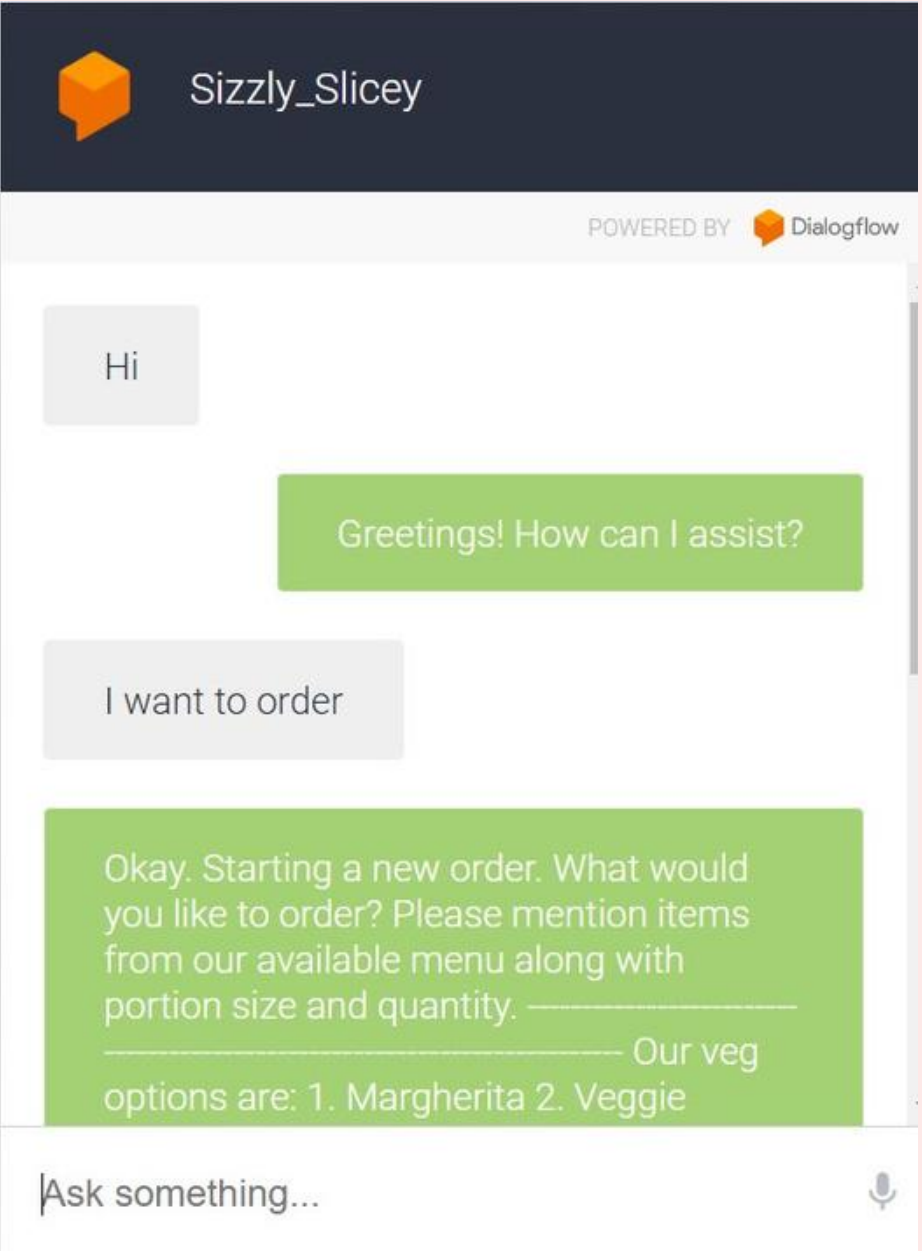
Sprint 2 Deliverables:



Add to Cart

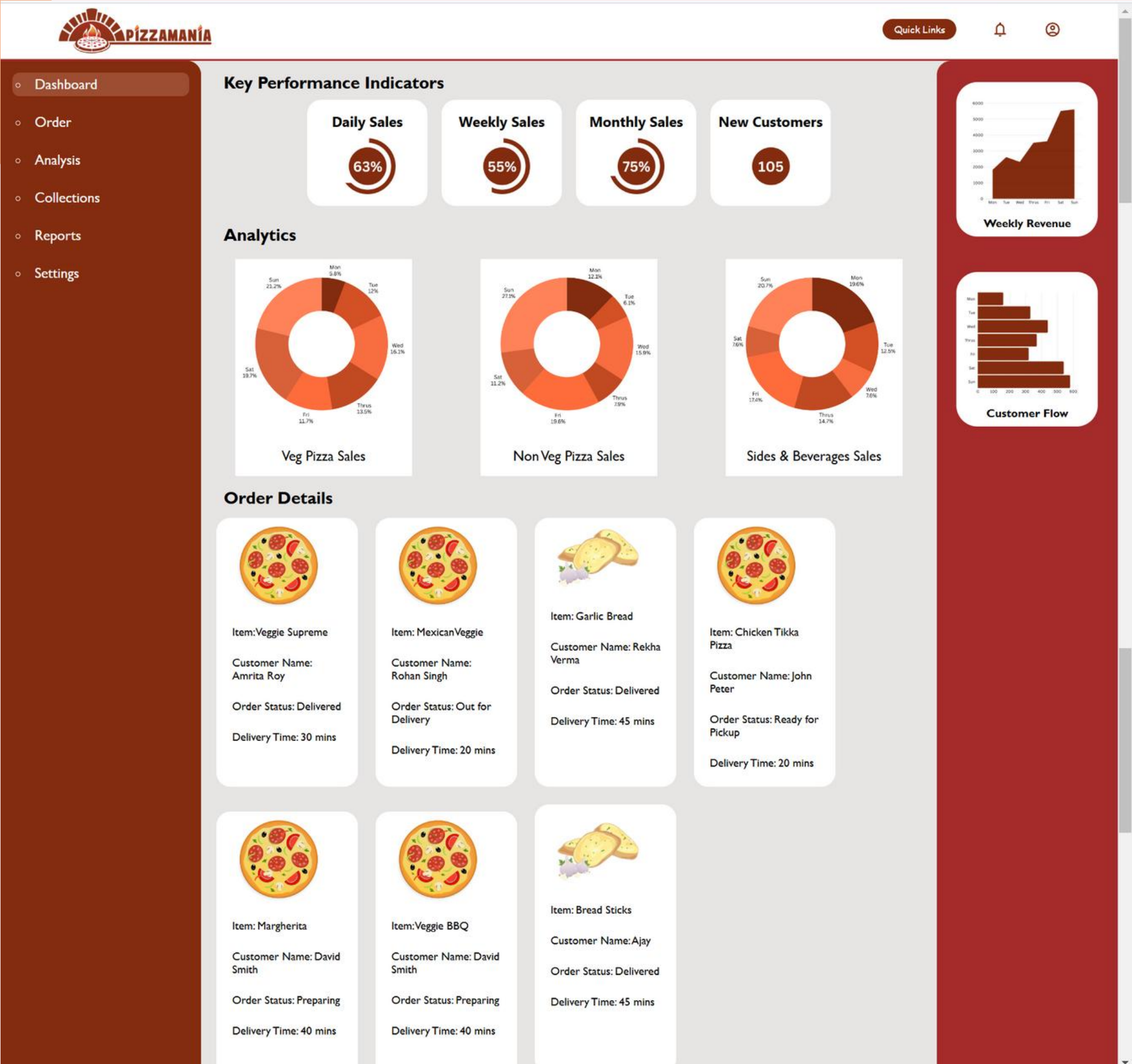


Delete items from Cart



Chatbot Help-Assistant

Sprint 2 Deliverables:



Update Menu

Item Name:

Description:

Price:

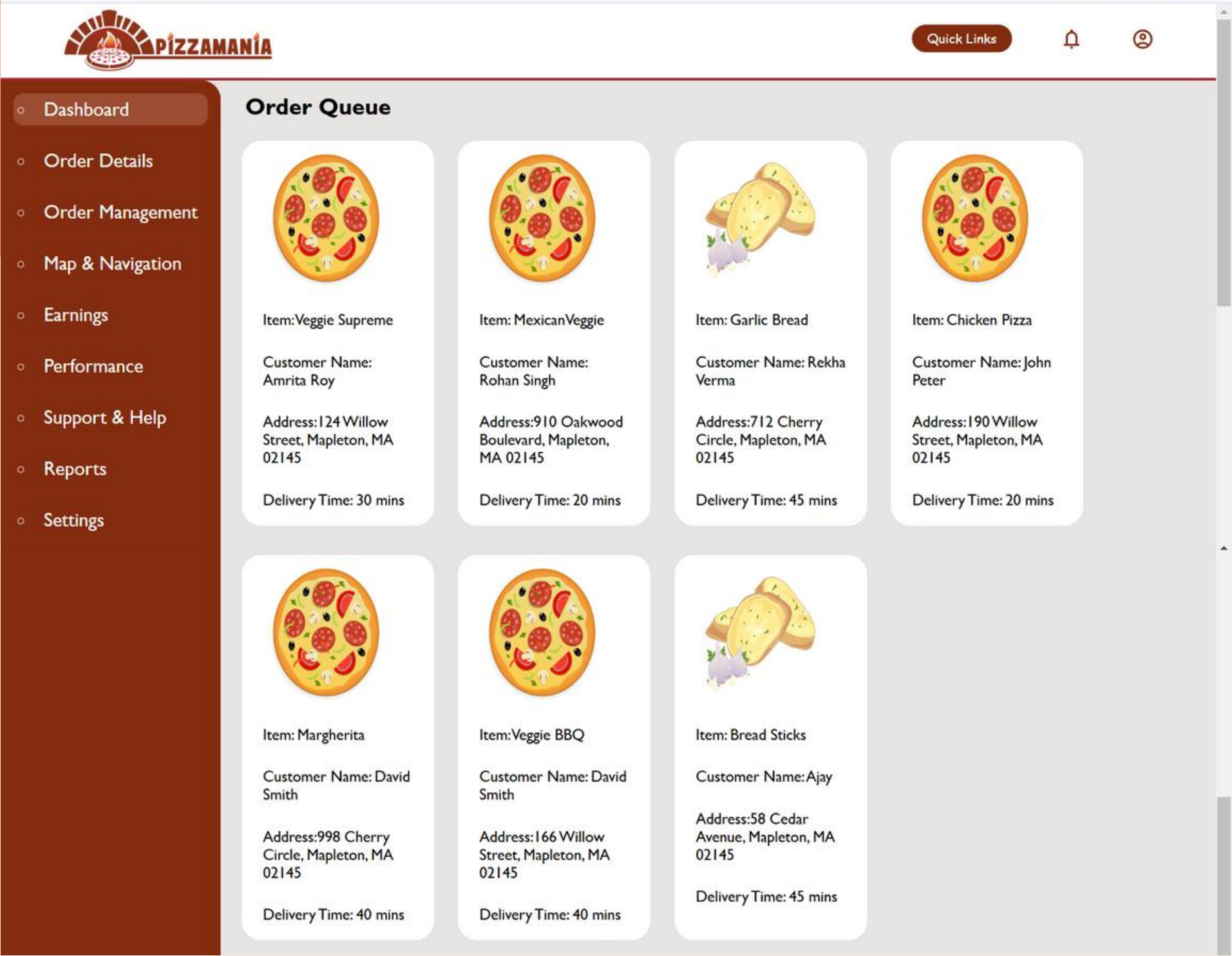
Category:

Update Menu

Way to add new items (Admin's side)

Admin Dashboard with Order Details & Sales Analysis

Sprint 2 Deliverables:



Delivery Agent Dashboard with Performance Metrics

Planning Poker Estimation by Scrum Team

Poker Estimation																								
SL NO	Name of Scrum Members (Scrum Master & Developmental Team)	Roushan			Tideesha			Debarati			Alivia			Koyelisha			Adrija			Shyantn			Merged Estimate of Each User Story	
	DESCRIPTION	R1	R2	R3	R1	R2	R3	R1	R2	R3	R1	R2	R3	R1	R2	R3	R1	R2	R3	R1	R2	R3		
1	Admin,Customer,Delivery boy should be able to login	55	55	34	21	34	34	21	21	34	89	55	34	21	34	34	21	55	34	21	34	34	34	
2	New user should be able to create new account	34	21	21	34	21	21	13	21	21	55	34	21	21	34	21	34	21	21	34	21	21	21	
3	User should be able to Browse pizza and sides in Menu	21	34	55	13	21	55	21	34	55	89	89	55	89	89	55	89	55	55	89	89	55	55	
4	User should be able to sort pizza by price	13	13	8	2	3	8	3	3	8	21	13	8	21	13	8	13	13	8	8	8	8	8	
5	User should be able to customize pizza by choosing different toppings and sizes	13	21	21	13	13	21	13	21	21	34	21	21	34	21	21	13	13	21	34	21	21	21	
6	User should be able to add items to cart	8	8	13	5	8	13	5	8	13	21	13	13	21	13	13	21	21	13	21	21	13	13	
7	User should be able to view cart summary	2	5	5	2	3	5	3	5	5	8	5	5	5	5	5	8	5	5	8	5	5	5	
8	User should be able to update cart items	8	8	8	5	8	8	5	8	8	13	8	8	5	5	8	13	5	8	13	13	8	8	
9	User should be able to apply coupon code	3	3	2	1	2	2	3	2	2	5	3	2	5	2	2	3	2	2	3	2	2	2	
10	User should be able to select payment method (Cash or Online) and make payment	55	55	89	34	55	89	55	55	89	89	89	89	55	55	89	89	89	89	89	89	89	89	
11	User should be able to track order in real time **	21	34	34	34	34	34	21	21	34	55	55	34	55	34	34	21	21	34	55	55	34	34	
12	User should be able to view pizza outlet's location and picture	1	1	2	1	2	2	2	2	2	3	2	2	2	2	2	2	2	2	2	1	2	2	
13	User should be able to search for existing public reviews	1	1	2	1	2	2	2	2	2	3	2	2	2	2	2	3	3	2	2	1	2	2	
14	User should be able to receive order confirmation through SMS	3	5	5	3	3	5	3	5	5	5	5	5	2	3	5	5	5	5	2	3	5	5	
15	User should be able to cancel order	21	13	13	21	21	13	21	21	13	34	21	13	21	21	13	21	21	13	21	13	13	13	
16	User should be able to receive cancellation confirmation through SMS	3	5	5	3	3	5	3	5	5	5	5	5	2	3	5	5	5	5	2	3	5	5	
17	User should get refund for the cancellation via online banking	21	21	21	34	21	21	34	21	21	34	34	21	34	21	21	34	34	21	21	21	21	21	
18	User should be able to receive the details of the delivery agent	8	5	8	5	5	8	3	5	8	5	5	8	5	5	8	5	5	8	3	5	8	8	
19	User should be able to update their personal information	1	2	2	5	3	2	3	2	2	1	1	2	2	2	2	1	2	2	1	1	2	2	
20	User should be able to update the delivery location	2	3	3	5	3	3	3	3	3	2	2	3	3	3	3	2	2	3	2	2	3	3	
21	User should be able to rate & review pizza	3	5	5	3	3	5	3	5	5	5	5	5	3	5	5	3	5	5	3	3	5	5	
22	User should be able to rate the delivery experience	3	5	5	3	3	5	3	5	5	5	5	5	3	5	5	5	5	5	3	3	5	5	
23	User should be able to save favourite orders	2	2	3	5	5	3	3	3	3	1	2	3	3	3	3	2	2	3	2	2	3	3	
24	User should be able to get recommendation of new pizza flavours based on past purchases	55	34	34	55	55	34	55	55	34	55	34	34	34	55	34	55	55	34	55	34	34	34	
25	User should be able to receive assistance for resolving order related issues through chatbot	55	34	34	55	55	34	55	55	34	55	34	34	34	55	34	34	34	34	55	34	34	34	
26	Admin should be able to update Menu items	13	13	21	34	21	21	34	21	21	21	21	21	34	34	21	34	21	21	34	21	21	21	
27	Admin should be able to adjust item pricing	5	8	8	5	5	8	5	8	8	5	8	8	8	8	8	8	8	8	3	5	8	8	
28	Admin should be able to view users' order	3	5	5	3	3	5	3	5	5	3	5	5	5	5	5	5	5	5	2	3	5	5	
29	Admin should be able to see the customer review	2	3	3	2	2	3	2	3	3	2	3	3	3	3	3	3	3	3	2	2	3	3	
30	Admin should be able to get summary sales report	34	55	55	89	55	55	55	55	55	34	34	55	89	55	55	34	55	55	34	55	55	55	
31	Admin should be able to see the current order status	34	55	55	89	55	55	55	55	55	34	34	55	89	55	55	34	55	55	34	55	55	55	
32	Delivery boy should be able to get assigned order information from admin via sms	34	21	21	34	34	21	34	21	21	34	21	21	34	34	21	34	21	21	34	34	21	21	
33	Admin should be able to view order payment status	55	34	55	34	34	55	34	55	55	89	55	55	55	55	55	89	55	55	89	89	55	55	
34	Delivery boy should be able to update delivery status	55	34	55	34	34	55	34	55	55	89	55	55	55	55	55	34	55	55	89	89	55	55	
	Total Story Points:																						710	

Product Backlog Snapshot

- ➔ The product backlog is designed to enhance a user-centric pizza ordering experience.
- ➔ High-priority features include login, account creation, and menu browsing functionalities.
- ➔ Prioritization is guided by critical user needs and core system functionality.
- ➔ Real-time order tracking and secure payment options are emphasized to ensure a seamless process.

Product backlog								
Feature ID	Feature	User Story ID	User Story Heading	Description	Acceptance Criteria	Priority	Estimate (Story Points)	Sprint
F001	User Authentication & Registration	S001	Admin, Customer, Delivery boy should be able to login	As an admin, customer, or delivery boy, I want to be able to log in to the PizzaMania platform, so that I can access my respective dashboard and perform my tasks.	User has access to a registration page. Requires name, email, phone, and password. Passwords are encrypted. Confirmation email sent after registration. User redirected to login page. The system should require essential details such as name, email, phone number, and password. Passwords should be encrypted for security. The user should receive a confirmation email upon successful registration. The user should be redirected to the login page after registration.	1	34	
		S002	New user should be able to create new account	As a new user, I want to be able to create an account on the PizzaMania platform, so that I can place orders and manage my preferences.	Separate login for admin, customer, and delivery boy. Redirect to unique dashboards after login. Validate credentials and show error for invalid attempts. User remains logged in until logout or session expiration. Each user role should be redirected to a unique dashboard upon successful login. The system should validate credentials and display an error message for invalid attempts. The user should remain logged in until they log out or the session expires.	2	21	
		S003	User should be able to Browse pizza and sides in Menu	As a user, I want to browse available pizzas and sides in the menu, so that I can view all options and choose items based on my preferences.	The menu should display available pizzas and sides with names, descriptions, images, and prices. Items should be organized by categories (e.g., Veg Pizzas, Non-Veg Pizzas, Sides). The user should be able to view details of each item. The menu should update dynamically to reflect any additions or deletions by the admin.	3	55	

Product Backlog Snapshot

F002	Menu & Recommendations	S004	User should be able to customize pizza by choosing different toppings and sizes	As a customer, I want to customise my pizza by selecting different toppings and sizes, so that I can create a pizza that meets my taste preferences.	The customer should see a list of available toppings and have the ability to select or deselect any combination. The customer should be able to choose from multiple pizza sizes (e.g., regular, medium, large) with the associated base price displayed. The customer should see a list of available crust types (e.g., Fresh Pan, Classic Hand Tossed, New Hand Tossed, Cheese Burst) along with their associated costs. The system should display real-time price adjustments based on the selected pizza size, crust type, and toppings. Once the pizza is customised, the customer should be able to add it to their cart. The cart should reflect the custom pizza with all chosen toppings and the correct size and preferred crust.	4	21	1
		S005	User should be able to get recommendation of new pizza flavours based on past purchases	As a customer, I should receive recommendations for new pizza flavors based on past purchases.	User receives flavor recommendations after logging in. Recommendations are based on user's purchase history. User can view detailed information about recommended flavors.	5	34	
		S006	User should be able to search for existing public reviews	As an end-user I need to be able to search existing public reviews of products or services so that I can make an informed decision before placing an order.	Users should be able to access public reviews directly from the homepage or a dedicated "Reviews" section, without needing a search bar or search functionality. Each review should display the reviewer's name (or an anonymous tag if privacy is needed), review date, star rating (if applicable), and the review text.	6	2	

Release Planning

➡ Feature Distribution: Each release groups related features, enhancing user functionality progressively.

Release #1

User Authentication & Registration:

- Login & Signup for New User, Admin & Delivery Agent

Menu & Recommendations:

- Listing, Sorting & Customization of All Food Offerings - pizza, Sides & Beverages
- Intelligent Recommendation & Public Reviews

Cart & Checkout:

- Adding & Updating Cart Items for Checkout
- Ability to View Cart Summary

Support & Admin Capabilities:

- User Support with Chatbot
- Admin control with all-in-one dashboard
- Delivery Info for Agent in one Dashboard

Release #2

Order & Delivery Management:

- Integration Of Payment gateway for Cashless transactions
- Delivery info sharing & order cancelation support to Customers
- Authorization to Admin for Order & Delivery Supervision
- Enhanced Support to Delivery Agents

Feedback & Notifications:

- Customer Satisfaction via Ratings
- SMS confirmation to User & Delivery Agent for Delivery/ Cancellation

Release #3

Refund & Account Management:

- Enabling User to update their info
- Aiding user with Refund support

Trust & Ease Of Delivery:

- Building User trust with Genuine Online reviews & pictures
- Ease of Ordering with Current Location

Admin Monitoring:

- Exclusive authority to Admin for Full Fledged Control & Supervision in Daily Operations & Management

Favorites & Discounts:

- Ease of Accessibility to users for repeating Orders & Avail discounts

***Core functionalities are Bold & Underlined**

Sprint Planning Calculation

→ Feature Distribution: Each release groups related features, enhancing user functionality progressively

No. of members = 8

No. of Hours worked by each member per day (minimum) = 2hrs

No. of Days in a Single Sprint = 10days

Total Working Hours in a Single Sprint = 8 members x 2hrs x 10days = 160 hours

No. of User Stories = 34

Total Estimated Story Points = 710 SP

No. of Sprints Required = $(710 \text{ SP} \times 1\text{hr}) / 160 \text{ hr} = 4.5 \sim 5 \text{ Sprints}$

Therefore, No. of Story Points to be covered in 5 Sprints = 710 SP

No. of Story Points to be covered in 1 Sprints = $(710/5) = 142 \text{ SP}$

Hence, required Velocity of the Scrum Team is “142 SP/Sprint”.

Sprint Planning

- Feature Prioritization: High-priority features placed in earlier sprints for quick impact.
- Effort Estimation: Story points assigned to match capacity, balancing sprint workloads.

Sprint #1

Admin, Customer, Delivery boy should be able to login

New user should be able to create new account

User should be able to Browse pizza and sides in Menu

User should be able to customize pizza by choosing different toppings and sizes

User should be able to get recommendation of new pizza flavours based on past purchases

User should be able to search for existing public reviews

User should be able to sort pizza by price

Sprint #2

User should be able to add items to cart

User should be able to view cart summary

User should be able to receive assistance for resolving order related issues through chatbot

Admin should be able to view users' order

Admin should be able to see Sales Statistics

Delivery boy should be able to get assigned order information

User should be able to update Cart Items (remove)

Sprint #3

User should be able to select payment method (Cash or Online) and make payment

User should be able to receive the details of the delivery agent

User should be able to cancel order

Admin should be able to update Menu items

Admin should be able to adjust item pricing

User should be able to rate & review pizza

User should be able to track order in real time

Sprint #4

Delivery boy should be able to update delivery status

User should be able to receive order confirmation through SMS

User should be able to receive cancellation confirmation through SMS

User should get refund for the cancellation via online banking

User should be able to view pizza outlet's location and picture

User should be able to update the delivery location

Admin should be able to see the current order status

Sprint #5

Admin should be able to view order payment status

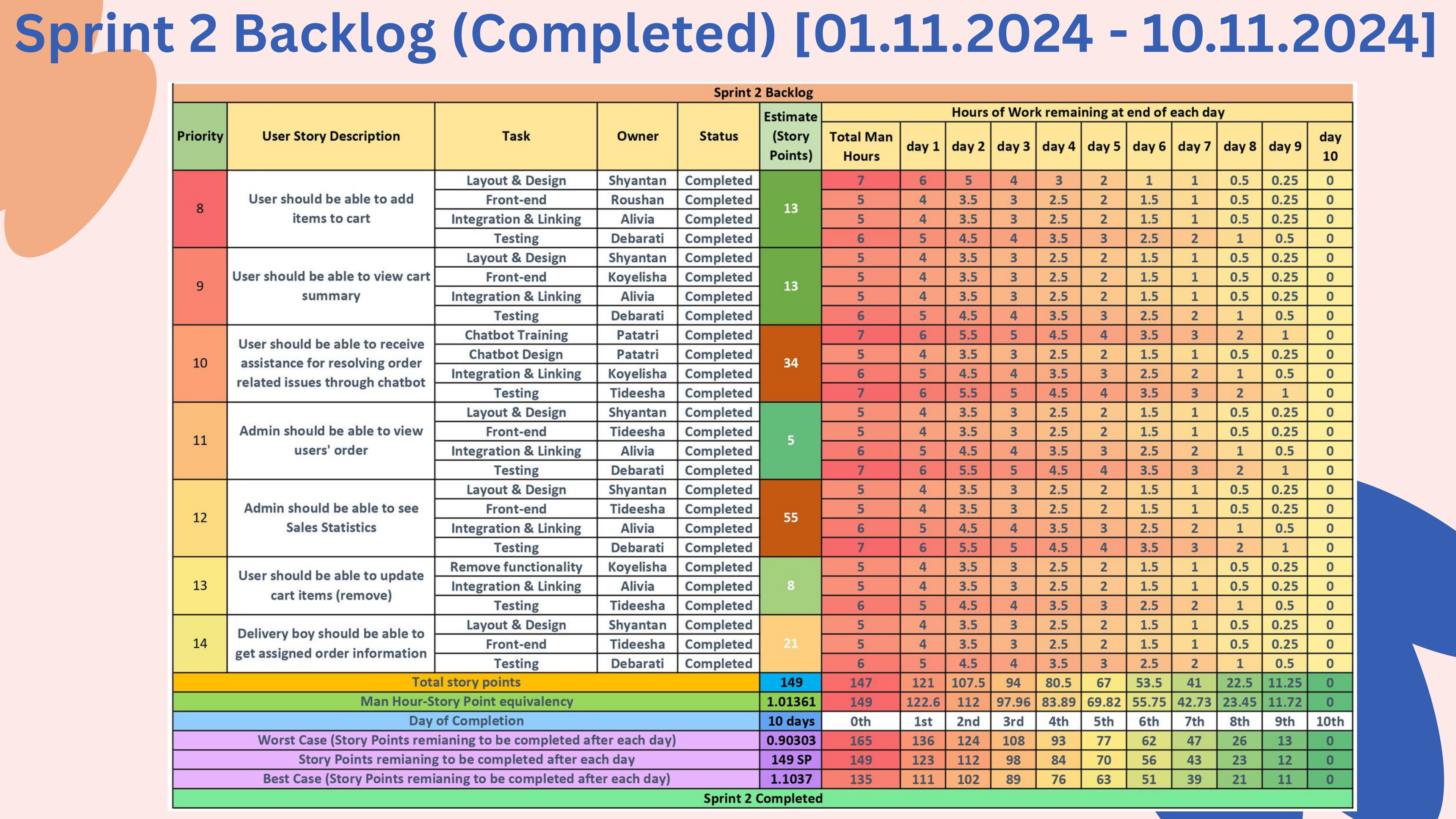
User should be able to rate the delivery experience

Admin should be able to see the customer review

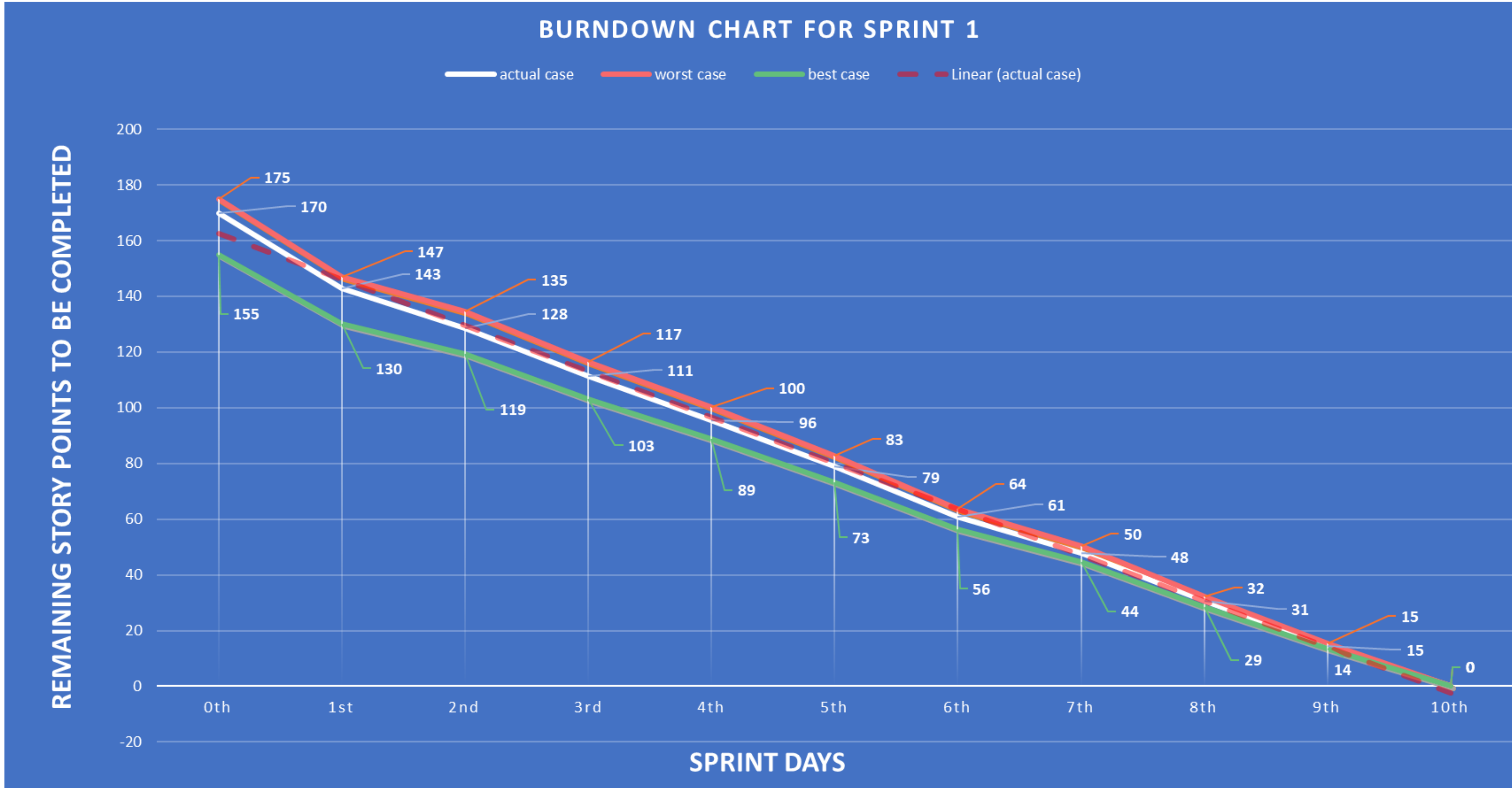
User should be able to save favourite orders

User should be able to apply coupon code

User should be able to update their personal information

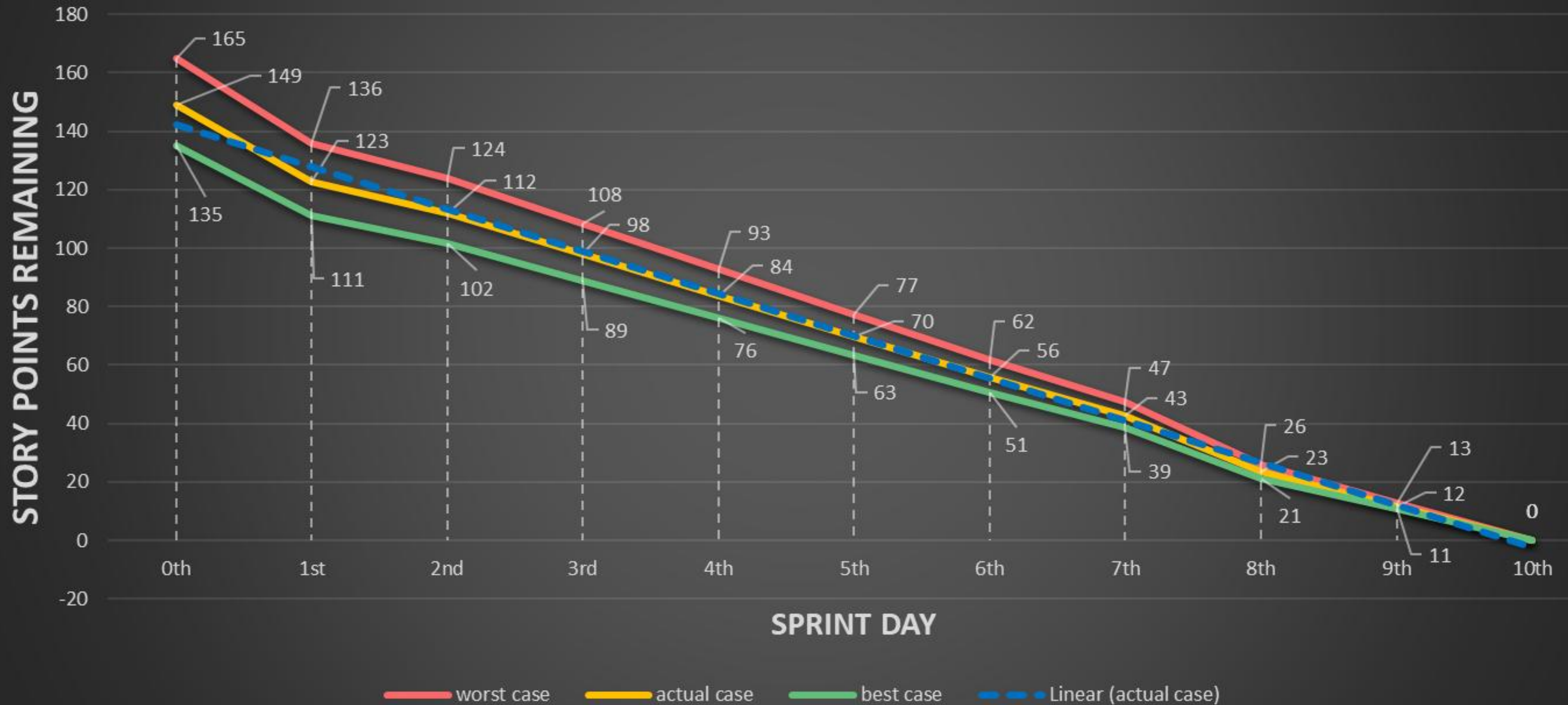


Burndown Chart for Sprint 1 Backlog (Completed)



Burndown Chart for Sprint 2 Backlog (Ongoing)

BURNDOWN CHART FOR SPRINT 2



Daily Standup

Issue Log>>



- ISSUE LOG SUMMARY
- Bug reports, performance issues, etc.
- Mean Resolution Time = 1

Issue ID	Description	Created By	Created On	Resolution Date	Resolution
IS01	Slow loading time on homepage	Tideesha	09-11-2024	10-11-2024	Optimized images and implemented lazy loading. Minified CSS and JavaScript files to reduce load time.
IS02	Header images resolution is too low, appears pixelated	Tideesha	08-11-2024	09-11-2024	Replaced with high-resolution images optimized for the web.
IS03	Transition on flex cards is not smooth	Tideesha	05-11-2024	06-11-2024	Optimized CSS transitions by reducing animated properties.
IS04	Pizza image thumbnails not loading	Tideesha	06-11-2024	07-11-2024	Confirmed image paths, tested across browsers, and added fallback images.
IS05	Slow loading time due to high-resolution images	Tideesha	04-11-2024	05-11-2024	Optimized images using compression and lazy loading techniques.
IS06	Inconsistent font styling between pizza names and descriptions	Tideesha	02-11-2024	03-11-2024	Standardized font styles for consistency across different browsers.
IS07	Price calculation errors when adding toppings	Tideesha	01-11-2024	02-11-2024	Retested and fixed logic for dynamic pricing on topping selection.
IS08	Incorrect sorting order (e.g., ascending instead of descending)	Tideesha	30-10-2024	31-10-2024	Verified sorting logic for all criteria and tested order functionality.
IS09	Sort button or icon lacks visual feedback	Tideesha	29/10/2024	30/10/2024	Added active and hover styles for sort buttons to enhance user feedback.

IS10	Animations cause significant delay on initial page load	Tideesha	23-10-2024	24-10-2024	Minimized file sizes for animation assets and implemented lazy loading for non-critical animations.
IS11	Comprehensive pizza dataset with detailed ingredient breakdowns not found	Adrija	26-10-2024	27-10-2024	Created a tailored dataset manually from restaurant menus for accurate recommendations.
IS12	Inconsistent ingredient names and duplicate entries causing errors in recommendation accuracy	Adrija	24-10-2024	25-10-2024	Standardized ingredient names and removed duplicates to enhance dataset uniformity.
IS13	Excessive memory usage and slow response times during KNN model training	Adrija	23-10-2024	24-10-2024	Applied PCA for dimensionality reduction and implemented an approximate nearest neighbors algorithm.
IS14	Features with larger numerical ranges skewing KNN model recommendations	Adrija	21-10-2024	22-10-2024	Standardized features to equalize their influence in distance calculations for balanced recommendations.
IS15	Layout not responsive on different screen sizes for browsing pizza and sides	Shyantana	20-10-2024	25-10-2024	Implemented CSS media queries for responsive design.
IS16	Missing pizza images causing layout misalignment	Shyantana	23-10-2024	28-10-2024	Added placeholder images to fix layout issues.

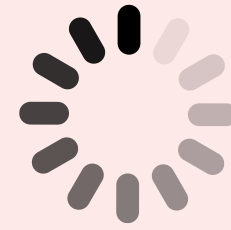
Kanban Board



To Do

Sprint 4 (order confirmation, refund , update location)

Sprint 5 (view payment status , coupon code , update personal information)



In Progress

Sprint 3 (payment method, cancel order ,update menu items, rate -review)



Done

Sprint 1 (Multi-role login,New user creation, menu browsing and customization , recoomendations based on past orders)

Sprint 2 (add to cart, view cart summary , chatbot, sales report)



Sprint 1 and 2 Restrospective

WHAT WENT WELL	WHAT WENT WRONG	WHAT NEED TO BE IMPROVED
Cart and Order Functionality: Users could add items to the cart and view a clear summary, enhancing the shopping experience.	Cart User Interface: The cart could be streamlined to make adding, removing, and updating items more intuitive.	Enhance Cart and Checkout Flow: Simplify cart updates, make checkout smoother, and ensure responsive loading.
Role-Specific Access: Admins could view orders and sales statistics, while delivery personnel received order assignments effectively.	Sales Analytics Detail: Sales statistics could include more granular insights, such as date filters and product trends.	Expand Sales Analytics Features: Add filters, visual graphs, and report-export options for more detailed admin insights.
Role-Specific Access: Admins could view orders and sales statistics, while delivery personnel received order assignments effectively.	Performance on High-Traffic Pages: The cart, checkout, and chat support pages could benefit from performance optimizations to reduce loading times.	Optimize Performance on Key Pages: Prioritize performance improvements for the cart, order summary, and chat support to handle more traffic.
Recommendation Precision: The recommendation engine could be refined to give more relevant suggestions based on individual user preferences.	Recommendation Precision: The recommendation engine could be refined to give more relevant suggestions based on individual user preferences.	Refine Recommendation Algorithm: Incorporate user purchase history and feedback to make pizza recommendations more personalized.

Conclusion

The conclusion of project management is not merely the end of a project but also an opportunity to glean valuable insights, celebrate accomplishments, and pave the way for future endeavors.

Thank everyone who contributed to this project.

Thank
you

